

NGUYEN ANH TUONG

SOFTWARE ENGINEER (ML / AI FOCUS)

SUMMARY

Final-year Computer Science student with strong hands-on experience in building and deploying production AI systems through internships at multinational companies, including Intel Corporation. Seeking Junior / Entry-level Machine Learning Engineer or Software Engineer roles with a focus on computer vision and applied ML systems.

CONTACT

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ACADEMIC HISTORY

VNU-HCMC University Of Science | 2022 - 2026

B.S in Software Engineering (Computer Science – Data Science)

- GPA: 3.8/4.0 (Major GPA: 4.0)

TECHNICAL SKILLS

Programming & CS Fundamentals - Python, C++, SQL, Data Structures & Algorithms, OOP

Machine Learning - PyTorch, TensorFlow / Keras, Scikit-learn, Supervised & Unsupervised Learning, Model Evaluation, Hyperparameter Tuning, Information Retrieval (BM25, Dense Embeddings, FAISS, Vector Similarity Search)

Computer Vision & NLP - Object Detection, Image Segmentation, OpenCV, NLP Pipelines, Retrieval-Augmented Generation (RAG)

MLOps & Deployment - Model Serving, Inference Optimization, Experiment Tracking, CI/CD, Docker, Kubernetes

Backend & Systems & Testing - RESTful APIs, FastAPI, Linux, Basic System Design, Locust

Cloud & Platforms - Google Cloud Platform (GCP), Microsoft Azure, ONNX, Intel OpenVINO, Intel Geti

WORK EXPERIENCE

AI Engineer Intern | 11/2025 - Present

Intel Corporation

- Contributed to the design and deployment of an end-to-end computer vision system for semiconductor surface defect detection, automating manual inspection workflows.
- Implemented object detection and segmentation models for high-resolution chip imagery, addressing fine-grained defects and class imbalance.
- Built and optimized production-grade inference pipelines under strict latency and throughput constraints, improving inspection efficiency.
- Worked on integrating training, evaluation, and deployment workflows into internal MLOps platforms for scalable and reproducible production rollout.
- Converted and optimized models for CPU-based inference using ONNX and Intel OpenVINO.

AI / Machine Learning Engineer Entry Level | 07/2025 - 11/2025

AI / Machine Learning Engineer Intern | 03/2025 - 07/2025

Seedcom Food Co.

- Built and maintained production-grade AI services for search, recommendation, and conversational systems supporting 50,000+ daily active users.
- Designed end-to-end pipelines covering data ingestion, model training, inference services, CI/CD, monitoring, and failure recovery.
- Optimized system latency and scalability, consistently maintaining sub-300 ms inference latency under real user traffic.
- Worked closely with product and backend teams to translate business requirements into robust AI-driven software components.

SELECTED PROJECTS

I - Intelligent Search & Keyword Suggestion Platform

(Deployed in production across multiple e-commerce websites (kingfoodmart, sieuthisi) and mobile applications (onelife))

- Architected a hybrid retrieval system combining BM25, dense embeddings, FAISS, and Trie-based prefix completion for low-latency query suggestion.
- Implemented semantic ranking using BERT-based rerankers, NER, and graph-based knowledge representations.
- Engineered retrieval and ranking services achieving ~80 ms retrieval and ~200 ms end-to-end validation.
- Improved search relevance by 2.5× while reducing overall latency by 40%.
- **Delivered measurable business outcomes:** 16% Conversion Rate, 67% CTR, and 58% Add-to-Cart ratio.

II - OneChef – Personalized Conversational AI System

(Deployed in production across mobile applications (onelife))

- Built and deployed a production conversational AI system for intent understanding, knowledge retrieval, and personalized recommendations.
- Designed an end-to-end RAG pipeline integrating real-time retrieval with LLMs for context-aware, grounded responses.
- Implemented personalization and multi-turn context management using user profiles, embeddings, and intent signals.
- Optimized inference, caching, and prompt orchestration to improve reliability and latency in production.
- **Impact:** +35% user engagement, +22% response accuracy, ~30% reduction in end-to-end latency.

PUBLICATIONS

VNU-HCMC University Of Science | Conference: ACIIDS | 2026 | [Github](#) | [Paper](#)

AutoViVQA: A Large-Scale, Automatically Constructed Dataset for Vietnamese Visual Question Answering.

- Role: First Author.
- Objective: To automatically construct a large-scale Vietnamese VQA dataset using LLM-based multimodal reasoning, with emphasis on reasoning consistency, cultural relevance, and fairness-aware data balancing.

VNU-HCMC University Of Science | Ongoing work | 2026 | [Github](#)

A comprehensive Vietnamese Visual Question Answering system with state-of-the-art features

- Role: First Author, Research Engineer.
- Objective: To design an end-to-end Vietnamese Visual Question Answering system that integrates multimodal representation learning, mixture-of-experts architectures, and retrieval-augmented reasoning, enabling scalable, culturally relevant, and high-quality VQA data generation and model training.

HCMC Van Lang University | 2025 | [Github](#)

A multi-model pipeline for 3D brain tumor segmentation (BraTS), combining SAM2, UMamba3DUNet, and a Mixture of Experts (MoE) gating network.

- Role: Research Engineer.
- Objective: To develop a scalable 3D medical image segmentation system by combining foundation-model representations and domain-specific volumetric models through a mixture-of-experts architecture, enabling accurate boundary detection, deep 3D contextual reasoning, and efficient expert utilization.

CERTIFICATION

IBM AI Engineering Professional Certificate, IBM Data Science Professional Certificate, Google Project Management: Professional Certificate